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Electron Attachment

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October 15, 2024

Abstract

This experiment deals with the resonant scattering of CCl_4 and SF_6 molecules with free electrons. A hemispherical electron monochromator connected to a quadrupole mass spectrometer is used to analyze the dissociative electron attachment of the molecules. First, the mass spectrometer is calibrated utilizing a known Cl^- resonance located at 0 eV. A further Cl^- resonance, as well as a SF_6^- and SF_5^- resonance is recorded and corrected using the previous calibration measurement. Comparing to literature values, we notice significant deviation in the 0.94 eV Cl^- and 0.38 eV SF_5^- resonance. A possible explanation is that there are multiple scattering processes at play which we can not distinguish between, that alter the observed distribution.

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Contents

0 Introduction

Mass spectrometry is an important experimental technique that offers many possibilities to research the structure of matter on a molecular level. The identification of different substances is possible by measuring the mass-to-charge ratio of ions. This not only allows to gather information of the mass distribution of substances, but also provides knowledge about molecular structures, chemical bondings and also isotopic distribution. Thus, the technique of mass spectrometry is applied in many fields of science, such as physics, chemistry, biology but also medical researches.

In this experiment, a hemispherical electron monochromator is used in combination with a quadrupole mass spectrometer. The monochromator consists of many electromagnetic lenses used to guide and focus an electron beam, which is necessary to study collisions between free electrons and molecules. Using this setup, the dissociative electron attachment is studied. In this process electrons scatter resonantly with molecules. After the scattering process, the molecule decays into a neutral and a negatively charged compound. With the quadrupole mass spectrometer, the mass and thereby the species of ion can be identified. For this experiment, two molecules are analyzed, CCl_4 and SF_6. Tetrachlormethan is used to calibrate the energy scale. This can be realized, as a peak at 0 eV and another peak are known. Using this knowledge, a zero offset can be determined. After this calibration, the two compounds of SF_6, namely SF_6^- and SF_5^-, are analyzed. Another part is the schematic analysis of a potential-energy-diagram for SF_6 and SF_6^- by approximating the potential with a Morse-potential.

The basic structure of this report will be as follows. The first section provides the necessary theoretical background of electron scattering and the dissociative electron attachment, which is a essential element of the experiment. The next chapter describes the experimental setup with a hemispherical electron monochromator and a quadrupole mass spectrometer. Afterwards, the execution and the experimental settings are provided. The fourth section presents the results, including plots, followed by a short summary and discussion.

1 Theory

In this chapter, the necessary theoretical background of the experiment is described, with dissociative electron attachment as main focus. The content of the theoretical part refers to the script of the experiment [1].

Scattering processes of electrons with molecules can be classified in direct and resonant or indirect scattering. While direct scattering happens fast, resonant scattering needs a longer time. Here, the incident electron's kinetic energy equals a resonant state of the molecule, it then creates a temporary negative ion (TNI):

$\mathrm{M} + \mathrm{e}^- \rightarrow \mathrm{M}^{*-}$.

This excited complex removes its excess energy in one of four possible ways. In elastic scattering $\mathrm{M}^{*-} \rightarrow \mathrm{M} + \mathrm{e}^-$ the electron leaves the molecule, which remains in its original state. Whereas in inelastic scattering $\mathrm{M}^{*-} \rightarrow \mathrm{M}^* + \mathrm{e}^-$, the molecule is left in an excited state. These two reactions are summarized as autodetachment. Another option is the emission of radiation, in which case a photon is emitted by the negatively charged molecule: $\mathrm{M}^{*-} \rightarrow \mathrm{M}^- + \mathrm{h}\nu$. The fourth of the competitive processes is the dissociative electron attachment (DEA). Here, the molecule dissociates and leaves a neutral and a negatively charged fragment:

$\mathrm{M}^{*-} \rightarrow \mathrm{A} + \mathrm{B}^-$

The DEA process can be explained using a Born-Oppenheimer potential-energy-diagram, as shown in Figure 1. Here the y-axis shows the potential energy and the x-axis indicates the distance between the two nuclei of a diatomic molecule AB. Due to electron capture, the molecule's energy increases, which is represented as a vertical line upwards from the ground state of the molecule. This has to be in agreement with the Franck-Condon principle, as the distance between the two nuclei cannot change that rapidly following the electron capture. The created, unstable TNI can relax either through autodetachment or DEA. If the distance reaches a level above a critical distance R_c, only DEA can occur.

In the right graph in Figure 1, the ion yield can be seen, depending on the initial electron energy. The distribution is slightly asymmetric, as the yields resulting from electron attachment, without considering autodetachment, and from DEA, including autodetachment, are overlapping.

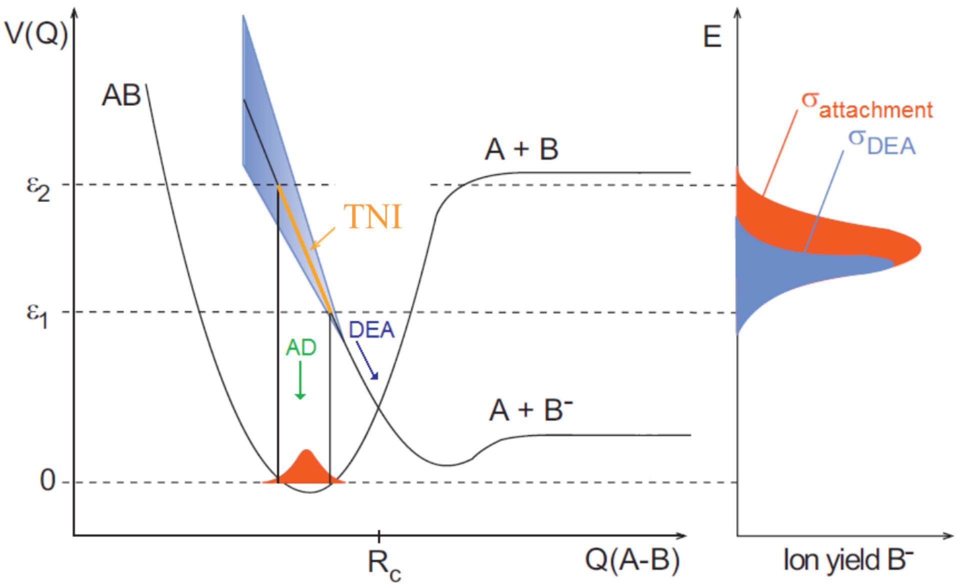

Figure 1: On the left, there is a Potential energy diagram of a diatomic molecule AB. On the right, the ion yield of B^- can be seen. Figure taken from [1].

An important property regarding the DEA is the adiabatic electron affinity AEA, which is the energy difference between the ground state of the neutral molecule and the ground state of the anion. The dissociation energy D_0(A-B) is the energy difference between the ground state of the neutral molecule and the asymptote, where the molecule is split in its compounds A+B.

To calibrate the setup, CCl_4 is used as a gauge molecule. This gas is very useful, as it reveals a peak at 0 eV in the spectrum. This peak makes it possible to determine the offset (from the peaks center) as well as the energy resolution (from the full width at half maximum) of the incident electron beam. The peak at 0 eV is a result of the fact, that even electrons with no kinetic energy can create an anion out of the ground state of the molecule, as one curve of the anion in the potential energy diagram is intersecting the ground state of the neutral molecule. Another peak is expected at 0.94 eV, resulting from another anionic potential path.

2 Experimental setup

The main element of the experimental setup is the Hemispherical electron monochromator (HEM), whose main components are an electron source, a set of lenses, a hemispherical energy selector, an ionization chamber and a Faraday cup. Another important part of the setup is a quadrupole mass spectrometer.

The first element is the electron source, a hairpin filament, through which a current of approximately 2.35 A passes. In our setup, 2.3 A are used. Thereby electrons are emitted by the filament with a broad energy distribution and a wide angular spread. A voltage of 60 to 100 V accelerates the electrons towards an anode. To focus and direct the electron beam properly, a set of electromagnetic lenses is used. These create electric fields, which guide and focus the electron beam like an optical lens manipulates a light beam. All lenses are supplied with voltages. These voltages are adjusted to gain maximum signal efficiency and additionally achieve a precise energy resolution. After the first set of lenses, the beam enters a hemispherical energy selector, made of two concentric hemispheres, held at two different voltages. Using the spherically symmetric field between the two surfaces, electrons with a certain energy can pass the selector, while others get absorbed by one of the hemispherical surfaces, as their trajectories get bend too much or too little. Before the beam enters the selector, it is adjusted by a deflector. After the selector, another set of lenses focus the beam further, before it enters the collision chamber, in which it ionizes gas molecules. There, neutrally charged gas molecules are inserted by a leak valve. The ions, which are created by collisions of molecules with electrons, are guided towards a quadrupole mass filter by a weak electric field. All excess electrons are collected by a Faraday cup, a plate connected to a picoamperemeter, which detects the intensity of incoming electrons. The quadrupole mass spectrometer, consisting of four parallel rods, separates ions after their mass to charge ratio. On two opposite rods, the voltages $\phi_{\pm} = \pm(U + V cos(\omega t))$ are inserted, where U represents the amplitude of the direct voltage, V is the amplitude radiofrequency voltage. In the spectrometer, ions are oscillating, but only the ones with a certain mass to charge ratio leave the spectrometers, the others get absorbed, as their oscillating amplitude is increasing too much to pass the quadrupole system. The detection of the ions is realized using a linear channel type secondary electron multiplier and a curved electron multiplier. To protect the electron beam from external magnetic fields, three pairs of Helmholtz-coils are located around the HEM.

3 Experimental execution

The first part of the experiment is to adjust the electromagnetic lenses by varying the input voltages. Thereby, the voltages L4, D1, D2, Ua, Ui, L5, 3a, 3b, 3c, A2, L6, L7, 4a, 4b, 4c and L8 are changed, the others remain untouched. The lenses should be optimized to maximize the electron current detected by the picoamperemeter connected with the Faraday cup. The final voltages used in the experiments can be seen in Table 1.

For the first measurement, the collision chamber is flooded with CCl_4. Gas molecules get ionized, and Cl^- ions are detected in the spectrometer. In order to analyze the Cl^- yield in the spectrometer, two measurements at mass 35 u were recorded. Afterwards, the mean of both measurements was computed. The measurement is done to calibrate the spectrometer. The measured pressure at this measurement is 5.2×10^{-7} mbar.

Then, a sample of SF_6 molecules is added instead of CCl_4. Here, the yield of different ions is analyzed, namely SF_6^- and SF_5^-. The yield for SF_6^- is measured two times at a mass of 146.6 u. The measured pressure at this time is 4.2 mbar. The yield for SF_5^- is determined five times at a mass of 127.7 u. The pressure here is 3.1 mbar.

Table 1: The voltages used in the experiment are listed with the corresponding label. Please refer to [1] for a detailed experimental setup with matching labels.

Lens	Voltage
1a	2.1 V
1b	−11.3 V
1c	−14.7 V
2a	−0.05 V
2b	0.00 V
2c	−0.03 V
3a	1.2 V
3b	−16.1 V
3c	−40.9 V
4a	2.3 V
4b	5.9 V
4c	36.1 V
L2	51.0 V

Lens	Voltage
L3	1.5 V
L4	10.5 V
L5	21.0 V
L6	−1.9 V
L7	0.00 V
L8	0.04 V
L10	0.00 V
A1	12.4 V
A2	69.8 V
A3	0.0 V
D1	13.6 V
D2	14.6 V
Anode	32.2 V

Lens	Voltage
Ie	−37.4 V
Da	−17.9 V
Di	22.7 V
Opt aus	76.0 V
Opt in	34.2 V
Ue	22.7 V
Uexmit	−1.9 V
Ua	−0.1 V
Ui	0.09 V
Uhk mit	2.2 V
SK	0.0 V
SK2	0.0 V
Ued	4.7 V

4 Results

We start by flooding the collision chamber with CCl_4 molecules at a pressure of $5.2 \cdot 10^{-7}$ mbar. Theory predicts a peak at 0 eV [1], which we will use to calibrate our x-Axis of further measurements. We choose the interval to include the 0 eV, as well as the neighboring peak, centered at 0.94 eV [1]. We sweep twice and average out the data to get a smoother signal. The collected data is depicted in Figure 2. We can identify two distinct peaks, to which Gaussian functions of the form $f(x) = A \cdot \exp\left(4\log(2)(x - x_0)^2/\mathrm{FWHM}^2\right)$ are fitted, with a scaling factor A, the center x_0 and the full width at half maximum FWHM as free parameters.

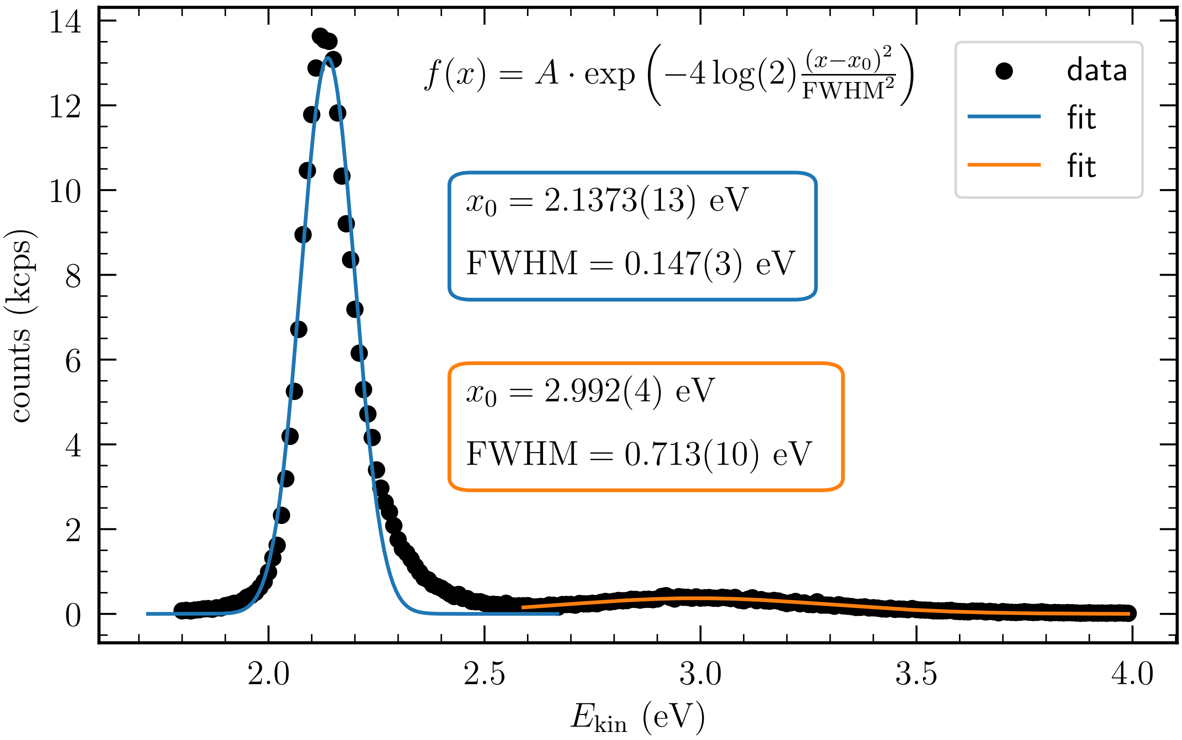

Figure 2: The recorded ionisation events of Cl^- are plotted against the kinetic energy of the incident electron beam. Two peaks can be identified in the data, to which Gaussian functions were fitted, represented by solid lines. From these we can extract the center and width of the peaks.

We start by analyzing the fit to the more prominent peak (solid blue line). If we identify this peak to be the 0 eV resonance, we can use the fitted center of the Gaussian $\varepsilon_1 = 2.1373(13)$ eV to calibrate subsequent measurements by offsetting them. The FWHM $= 0.147(3)$ eV of the Gaussian is directly linked to the energy spread of the incident electrons, which are distributed according to Boltzmann statistics.

We can now correct the energy of the second peak, resulting in an energy of $\varepsilon_2 = 0.854(4)$ eV with a spread of FWHM $= 0.713(10)$ eV. We can notice a significant difference to the literature value of 0.94 eV [1]. As can be seen in Figure 1, the ion yield distribution resulting from auto-detachment and dissociative electron attachment overlap. This in turn can cause the observed ion intensity peak to be shifted in energy compared to the expected value, since we are observing two different processes in the same energy range. The spread can be explained using the reflection principle, as the incident distribution is not not reflected along a straight line and therefore gets stretched out.

Now that we have conducted a calibration measurement we can analyze different molecules, in our case SF_6. We start by injecting SF_6 into the chamber at a pressure of 4.4 mbar and start looking at the resonant peak associated with SF_6^-. Again we sweep different kinetic energies and repeat this measurement to smooth the data. The data including a Gaussian fit can be seen in Figure 3.

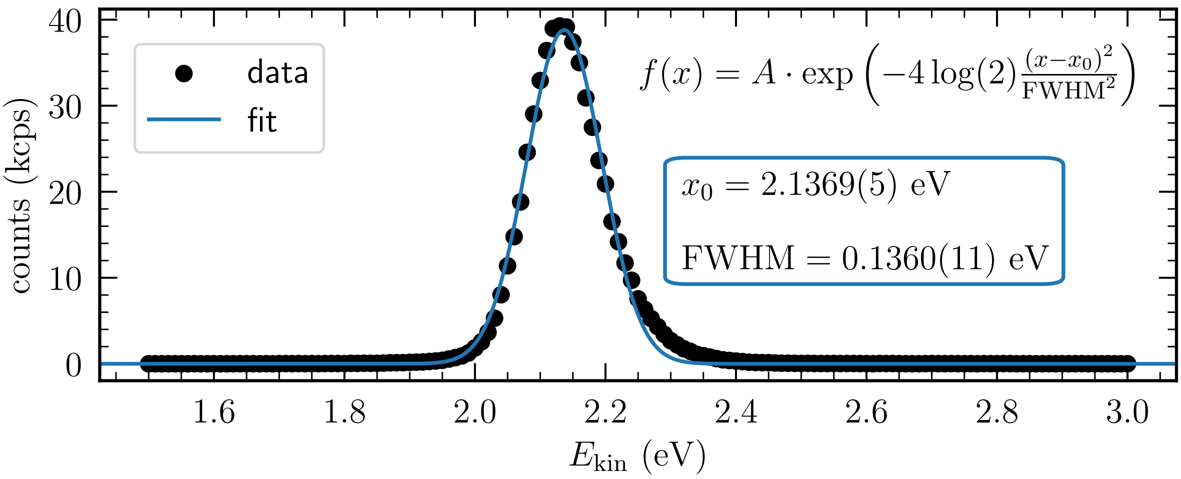

Figure 3: The recorded ionisation events of SF_6^- are plotted against the kinetic energy of the incident electron beam. A prominent peak can be identified in the data, to which a Gaussian function was fitted, represented by a solid line. From this we can extract the center and width of the peak.

From the fit we extract a resonance energy $\varepsilon_{SF_6^-} = 2.1369(5)$ eV, which after correction with the 0 eV Cl^- peak amounts to $-0.0005(14)$ eV. This result is consistent with the theoretical prediction of 0 eV [2]. The FWHM of the fit is comparable to the FWHM of the 0 eV Cl^- peak, since they both resemble the energy spread of the electron beam.

By tuning the mass spectrometer we can filter for SF_5^- anions. Here we repeated the energy sweep five times due a less distinct peak. In Figure 4 the data is plotted against the kinetic energy of the electrons, accompanied by a Gaussian fit.

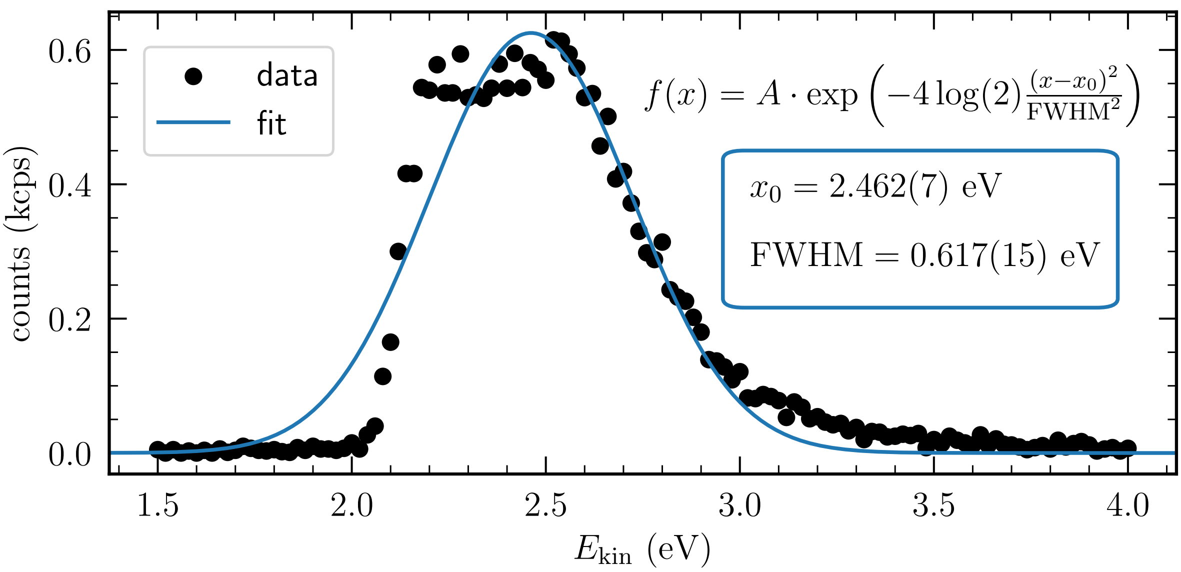

Figure 4: The recorded ionisation events of SF_5^- are plotted against the kinetic energy of the incident electron beam. A prominent peak can be identified in the data, to which a Gaussian function was fitted, represented by a solid line. From this we can extract the center and width of the peak.

At first sight, the data in Figure 4 seems to have a single asymmetric peak with a plateau at the center. When comparing to literature values [2] one can notice, that we expect two peaks in this energy range: one at 0.38 eV and another at 0.5(1) eV. The overlap of these two resonances is a possible explanation for the strange behaviour depicted in Figure 4. However this does not explain the corrected energy of the peak at $\varepsilon_{SF_5^-} = 0.324(7)$ eV, which lies below both resonances. As for Cl^-, we could explain this behaviour as the result of a mixing of underlying processes.

The next step is to draw a schematic potenital energy diagram of SF_6^- and SF_5^- to compare with the experimental results In order to plot such a diagram, some quantities have to be determined. The Morse potential $V_M(r)$ [3] is given by

$$V_M(r) = D_e[e^{-2a(r-R_0)} - 2e^{-a(r-R_0)}], \tag{1}$$

where D_e is the bond energy, R_0 the equilibrium distance and a the stiffness of the potential. We have given that $D_0(SF_5\text{-}F)$=4.35 eV and $D_0(SF_5^-\text{-}F)$=1.44 eV. Together with the zero-point energy of SF_6 and SF_6^-, 0.056 eV and 0.045 eV, respectively, D_e can be calculated for the anion as well as for the neutral molecule, as the sum of zero-point-energy and dissociation energy. The stiffness is calculated as

$$a = \omega_0 \sqrt{\frac{\bar{M}}{2D_e}}, \tag{2}$$

where $\bar{M}$ is the reduced mass of the molecule, in our case 16.527 u, and ω_0 is the fundamental frequency. This frequency is calculated from the zero-point-energy E_0 as

$$E_0 = \frac{\hbar\omega_0}{2}. \tag{3}$$

Additionally, the adiabative electron affinity AEA has to be considered to shift the potential curve of the anion downwards to a lower energy, which was done in Figure 5. It can be seen that the potential curves of SF_6 and SF_6^- cross each other at the ground state of the neutral molecule. This is the reason a 0 eV peak is expected in the spectrum, as also electrons with no kinetic energy are able to create an anion out of SF_6. In the schematic diagram, only one anionic potential curve is drawn, why only statements about this transition can be made. In reality, more than one curve exists for an anion. Additionally, the Morse potential is only an approximation.

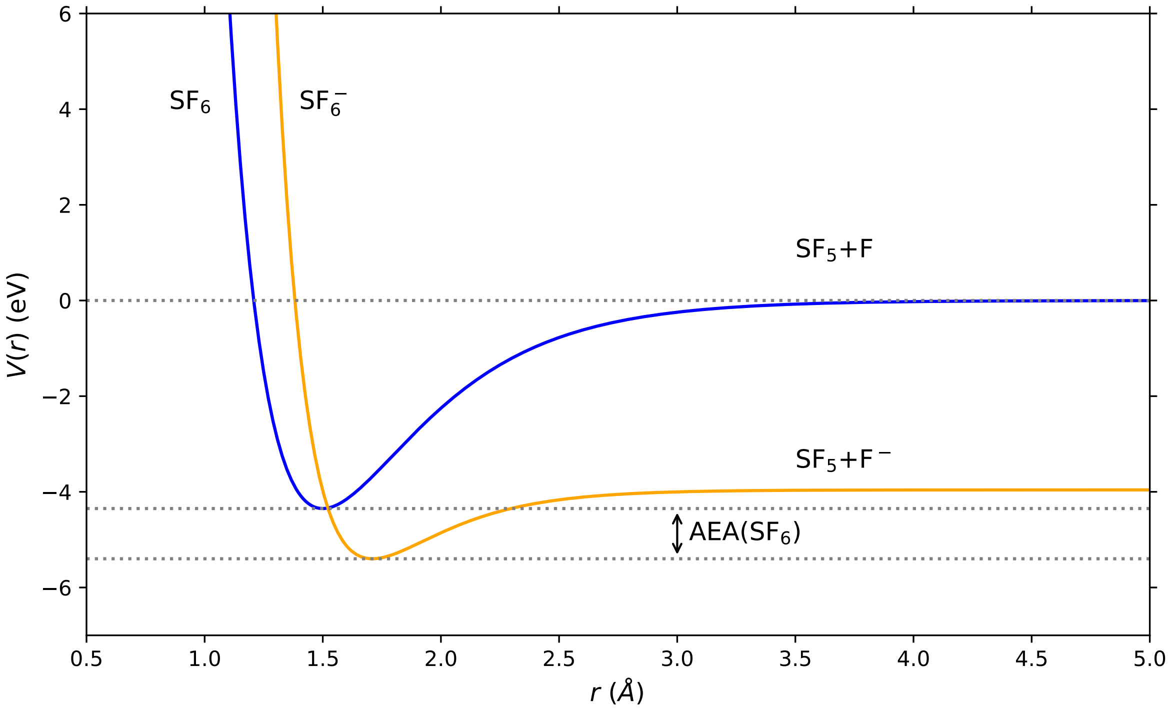

Figure 5: Schematically drawn Morse potential of SF_6 and SF_6^-.

5 Discussion

We start by optimizing the experimental setup to maximize the electron beam intensity and performing a calibration measurement with CCl_4. We successfully identify the 0 eV peak and use its center to offset all further measurements. We can further identify another peak at 0.8 eV, which is not in accordance with the literature value of 0.94 eV [1]. A possible explanation for this is that we actually observe the combination of autodetachment and dissociative electron attachment ion yield densities, leading to an offset peak.

After flushing the chamber and filling it with SF_6 we successfully identify a SF_6^- peak located at 0.0005(14) eV. This matches the literature value of 0 eV [2]. Next we scan for SF_5^- anions, which have expected peaks at 0.38 eV and 0.5(1) eV [2]. We can only measure one asymmetric and clipped peak, which indicates an overlap of the two resonances. A Gaussian profile was tentatively fitted with a center of 0.324(7) eV. This lies below both expected resonances and can therefore not be explained by a potential overlap. Again we can argue that the peak is shifted due to the autodetachment ion yield density. In the end, a schematic potential energy diagram was drawn. From that, it can be derived that a peak is expected at 0 eV, which agrees with the experimental result.

References

[1] Stephan Denifl. *Versuch FP-B-03 – Electron attachment*. Universität Innsbruck, Innsbruck, AT, 7.11.2022.

[2] L. G. Christophorou and J. K. Olthoff. "Electron attachment cross sections and negative ion states of SF611Dedicated to Professor Aleksandar Stamatovic on the occasion of his 60th birthday." In: *International Journal of Mass Spectrometry* 205.1 (2001). Low Energy Electron-Molecule Interactions (Stamatovic honor), pp. 27–41. ISSN: 1387-3806. URL: https://www.sciencedirect.com/science/article/pii/S1387380600002803.

[3] I. V. Hertel and C.-P. Schulz. *Atoms, molecules and optical physics*. Springer, 2014.

Erklärung

Hiermit versichern wir, dass der vorliegende Bericht selbständig verfasst wurde und alle notwendigen Quellen und Referenzen angegeben sind.

....	October 15, 2024
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